

PrivaCI: Privacy as Contextual Integrity

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In this special issue, we present an exciting “tasting sampler” of contextual integrity (CI) in application. Rooted in philosophy, CI offers an alternative to mainstream privacy definitions, which have struggled to address novel threats posed by powerful digital technologies and have yielded ineffective regulation.

Over time, the account of privacy as CI has gained traction across academic disciplines, inspiring research in computer science, human–computer interaction, social sciences, ethics, and law. It has extended into design and engineering practice, most recently, into artificial intelligence (AI) and large language model (LLM)-based agentic systems. Because application almost inevitably involves translating CI into the languages of different disciplines, it calls for far more than a straightforward projection of static and finished theory onto problems, issues, or systems in other spaces. In these formative years, CI is continuously being stress-tested and updated through disparate applications. New works, which may be controversial and contested, ultimately will point the way to productive development.

In each of the papers included in the issue, authors have identified challenges that expose shortcomings in mainstream privacy approaches, two which focus on regulatory approaches and a third that demonstrates how CI may be able to handle the seemingly intractable

challenges of remarkable advances in AI-augmented speech technologies.

In “Regulatory Contextual Integrity: Adaptively Regulating Privacy as Contextual Integrity,” Benthall and Sivan-Sevilla propose to embed the CI theory as part of the regulatory process to counter the predominant “notice and choice” paradigm.^{A1} They outline a procedure for an adaptive regulation that embodies the CI account of privacy, namely, ensuring the appropriate flow of information “in accordance with contextual norms, in service of general social goods.” The three-cycle procedure helps policymakers and regulators explore the legitimacy of established norms, enforce CI-based rules, and evaluate their effectiveness.

In “CA–CI: Integrating Contextual Integrity and the Capabilities Approach for Dignity Considerations in AI Governance,” Roemmich et al. examine the use of the CI theory in evaluating the fundamental rights impact assessments under the EU AI Act.^{A2} They propose to leverage the capabilities approach (CA) framework to establish harm thresholds in ensuring that, in addition to supporting contextual aims and purposes, information flows need to guarantee “a dignified life.”

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In “Differential Performance and Contextual Integrity in Speech Privacy,” Welch et al. propose a framework to examine privacy issues in modern speech technologies such as those used in home voice assistants and biometric authentication.^{A3} Their framework incorporates existing technical analyses of paralinguistic and extralinguistic factors in audio transmissions to the relevant CI parameter values to support a more comprehensive and robust privacy analysis of existing speech and audio technologies.

What unites these efforts, beyond their use of the five-tuple structure of CI-norms, is a commitment to evaluating data flows in terms of their support for contextual values and purposes. In day-to-day life, one wouldn’t need to evaluate data flows from first principles; each time, instead one relies on contextual norms, whose existence CI presupposes. Humans living social lives, acting in and across different social domains, absorb norms—what is and isn’t appropriate behavior through a variety of channels, some explicit, others implicit. Included among them are norms governing how and when data should and should not be shared from party to party. For CI, appropriate data flows, the touchstone of privacy as contextual integrity, are flows that conform with contextual informational norms or, absent applicable norms, are flows that balance diverse interests and effectively promote the ends, purposes, and values of disparate social domains such as health and education.

The rapid pace of sociotechnical change due, in large part, to LLM-based AI, challenges the status quo by introducing radically different data practices and upending privacy expectations. The CI community must meet these challenges in at least two ways. First, we should be prepared to describe these differences, rigorously, in terms of the five CI parameters; this enables us to detect shifts that are relevant to privacy and point to departures from expectations and norms. Second, once we have carefully described these changes, we should be prepared to identify impacts on the balance of stakeholder interests—winners and losers—and the extent to which contextual ends and values are served or disserved by these changes. The role for policy and governance is to demand sociotechnical systems that serve humanity and flourishing societies.

We are deeply grateful to the reviewers who took the time to provide invaluable feedback to us and to the authors. It was incredibly affirming to see the commitment and diligence of the members of our community, investing their time and effort to help advance the theory of contextual integrity. We would also like to thank all the people who helped make this

special issue possible: the authors who submitted their articles, the generous reviewers who provided essential input to our final selections, and the *IEEE Security & Privacy* editors and staff, especially journals production manager Erin Pacino and editors in chief Sean Peisert, Jeffrey Voas, and associate editor in Chief Mary Ellen Zurko, who oversaw the process for this special issue. Our reviewers Noah Apthorpe, Michael J. S. Beauvais, Elettra Bietti, Julia Bernd, Jake Chanenson, Catherine Dwyer, Severin Engelmann, Brett Frischmann, Matt Franchi, Jake Goldenfein, Ece Gumusel, Alexis Ingber, Kyle Jones, Emiram Kablo, Priya Kumar, Jae June Lee, Yafit Lev-Aretz, Nathan Malkin, Niloofar Miresghallah, Mainack Mondal, Elizabeth O’Neill, Elisa Orrù, Joel Reardon, Katie Shilton, Luke Stark, Thomas Streinz, Daniel Susser, Joseph Turow, Ine Van Zeeland, Joris van Hoboken, Meg Young, Elana Zeide, Michael Zimmer.

For further information and resources on contextual integrity please visit the <https://privaci.info> to check out the latest publications and join the community mailing list. ■

Appendix: Related Articles

- A1. S. Benthall and I. Sivan-Sevilla, “Regulatory contextual integrity: Adaptively regulating privacy as contextual integrity,” *IEEE Security Privacy*, vol. 24, no. 4, pp. 10–18, Jul./Aug. 2026, doi: [10.1109/MSEC.2026.3671696](https://doi.org/10.1109/MSEC.2026.3671696).
- A2. K. Roemmich, K. Martin, and F. Schaub, “CA-CI: Integrating contextual integrity and the capabilities approach for dignity considerations in AI governance,” *IEEE Security Privacy*, vol. 24, no. 4, pp. 19–29, Jul./Aug. 2026, doi: [10.1109/MSEC.2026.3654404](https://doi.org/10.1109/MSEC.2026.3654404).
- A3. P. Welch, N. Tomashenko, S. Das, and J. Williams, “Differential performance and contextual integrity in speech privacy,” *IEEE Security Privacy*, vol. 24, no. 4, pp. 30–39, Jul./Aug. 2026, doi: [10.1109/MSEC.2026.3679183](https://doi.org/10.1109/MSEC.2026.3679183).

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